Sci-Kitchen *Luxury Foods*

A PREDICTIVE SEGMENTATION STUDY

Ironhack Data Analytics Final Project | Claire Leyden

Context & Objectives

The Problem:

Optimising marketing spend based on standard digital metrics has failed to deliver significant ROI.

High website visits correlate poorly with sales, signaling casual browsing behavior rather than purchasing intent.

The Solution:

Using machine learning we can identify high-value demographic segments and divert our budget towards our most profitable customers.

Our Dataset

We profiled **2,240 customers** across **29 features,** encompassing demographics and past spending activity.

If we know who our customers are and how they behave can we predict conversion in our upcoming campaign?

Demographics included: Age, Marital status, Children

Behaviour included: Spending habits, Web visits

kaggle

Dataset: Customer Personality Analysis

Processing the Data

Data preparation

Cleaned rows with missing or implausible values. Resulted in final dataset containing **2,212** customers.

Feature Engineering

Merged & transformed feature columns. Converted categorical data to numeric and used standard scaling.

Dimension Reduction

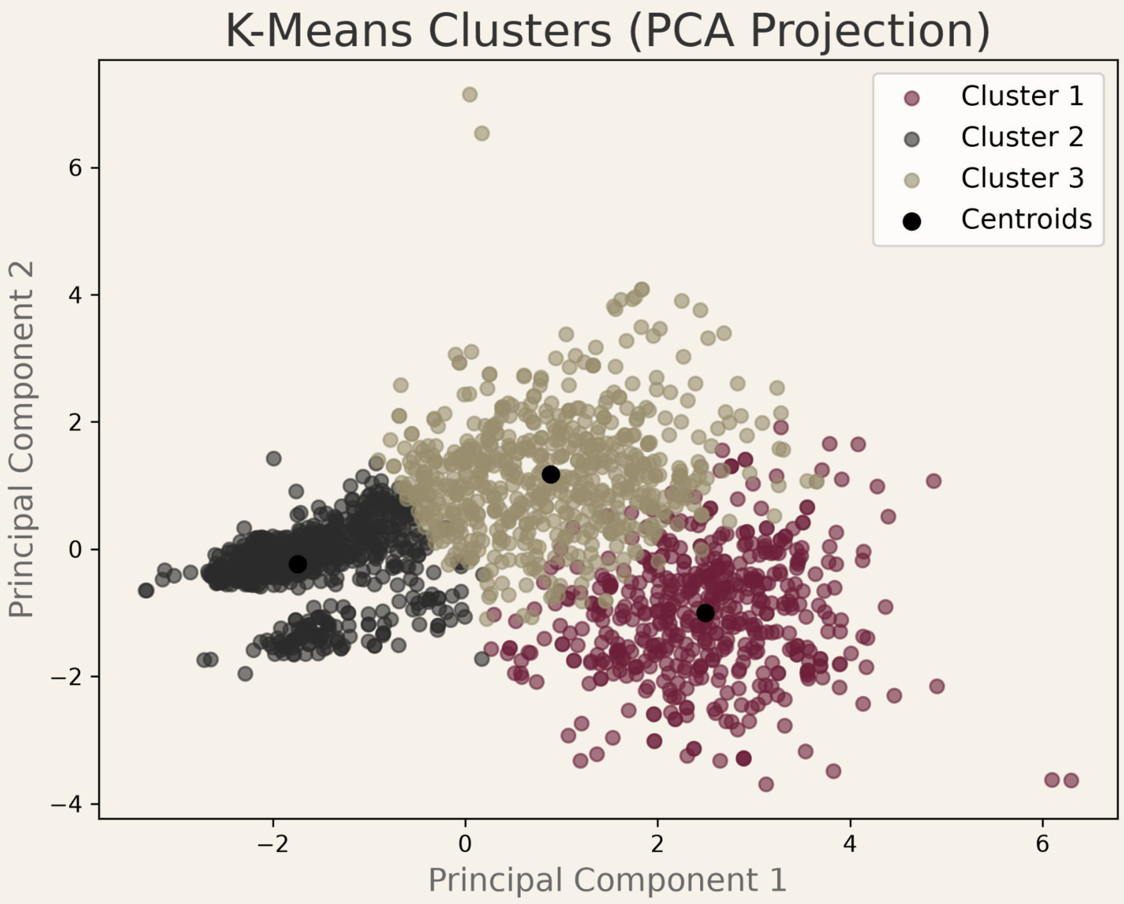
Used PCA to reduce 6 spending dimensions into 3 PCA components, capturing ~80% of the total variance.

Unsupervised Learning: K-Means Clustering

Combined 3 PCA components with 2 demographic and 3 spending characteristics to develop customer segments.

Result: 3 clusters emerged as best trade-off between silhouette score, inertia and marketing utility. 2 clusters did not allow for sufficient granularity.

The globular structure visible in PCA space confirms K Means as the appropriate clustering method.



3 Distinct Customer Profiles

Miguel, The Decisive Buyer

High income, no children.

Arrives informed. Converts fast.

Biggest spender.

Low digital footprint.

Our best customer.

Sofia, The Browser

Low income, two kids.

Spends a lot of time on website.

Buys little.

Our most visible customer. Not our most valuable.

Ana, The Family Spender

Mid income, children.

Engaged and considered.

Responds to the right offer.

Solid converter at 12% and our most reachable segment.

Segments Align with Exploratory Data Analysis

Behavioral Momentum

Significant relationship between previous behaviour and conversion..

4 Past Successes **90.9% Conv.**

3 Past Successes **79.5% Conv.**

2 Past Successes **51.9% Conv.**

1 Past Success **31.1% Conv.**

0 Past Successes **8.3% Conv.**

Demographics

Highly educated, childless customers convert.

26.6%

CONVERSION RATE (NO CHILDREN)

Drops to just **4.0%** for households with 3 children.

PhD Profile **21.0% Conv.**

Undergraduate **9.5% Conv.**

Cluster Performance

Segment identity correlated with conversion rate.

30.2%

DECISIVE BUYERS (CLUSTER 1)

Browsers **12.2% Conv.**

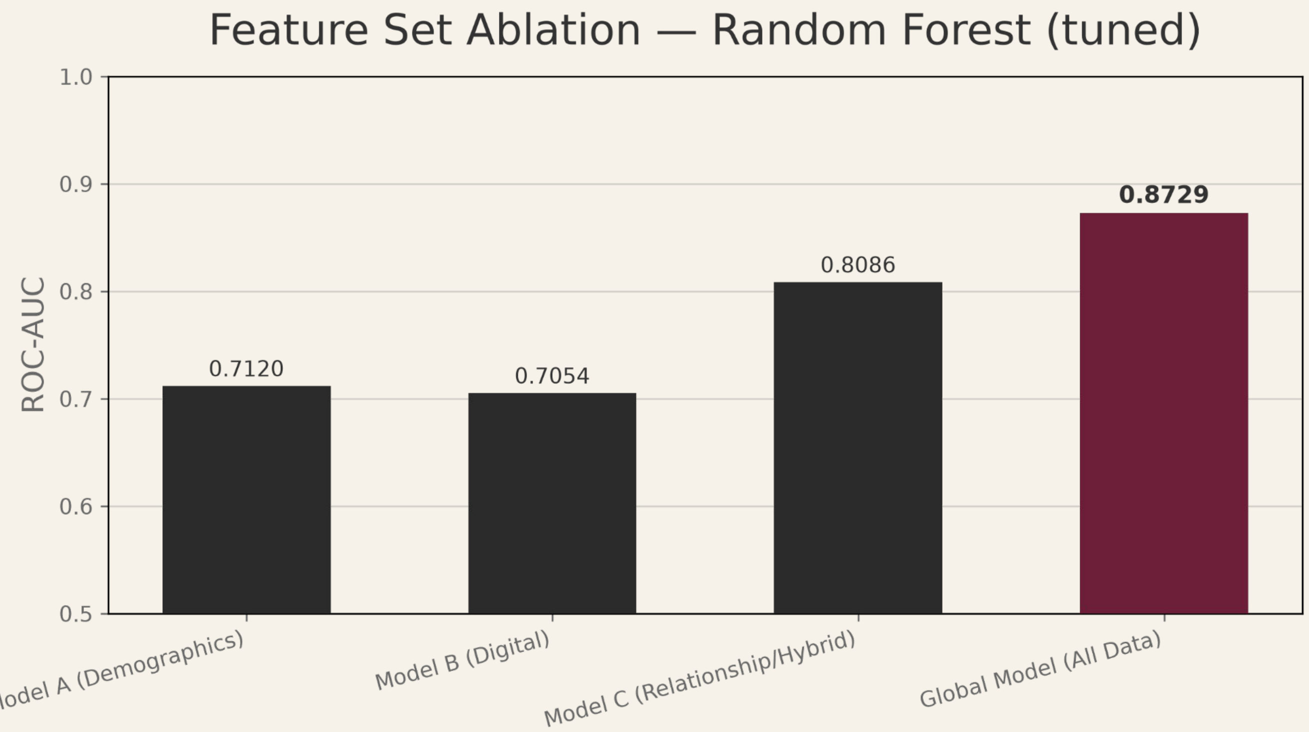
Family Spenders **9.4% Conv.**

Global Model Outperforms Subsets

Demographics and digital behaviour have similar predictive power in isolation.

Hybrid 3 CRM features plus cluster model outperforms both.

The global model adds only 6 points over the minimalist CRM model despite using three times as many features (13).



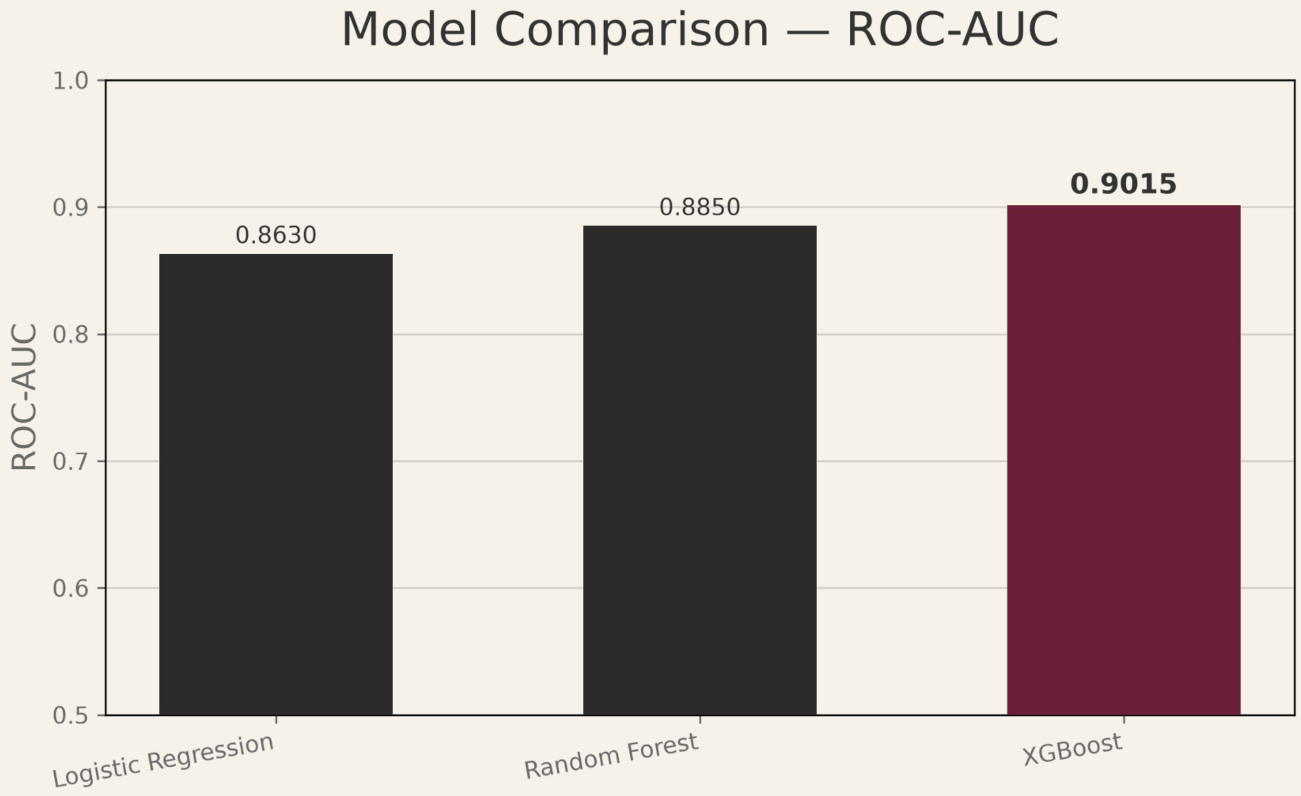
All models share the same hyperparameters, optimised for Model C.

XGBoost was Most Accurate Model

Hypertuned three classification models, Logistic Regression, Random Forest and XGBoost.

No significant difference in performance between Logistic Regression and Random Forest (DeLong's Test).

XGBoost significantly outperformed other models (Logistic Regression, p<0.01, Random Forest, p<0.05).

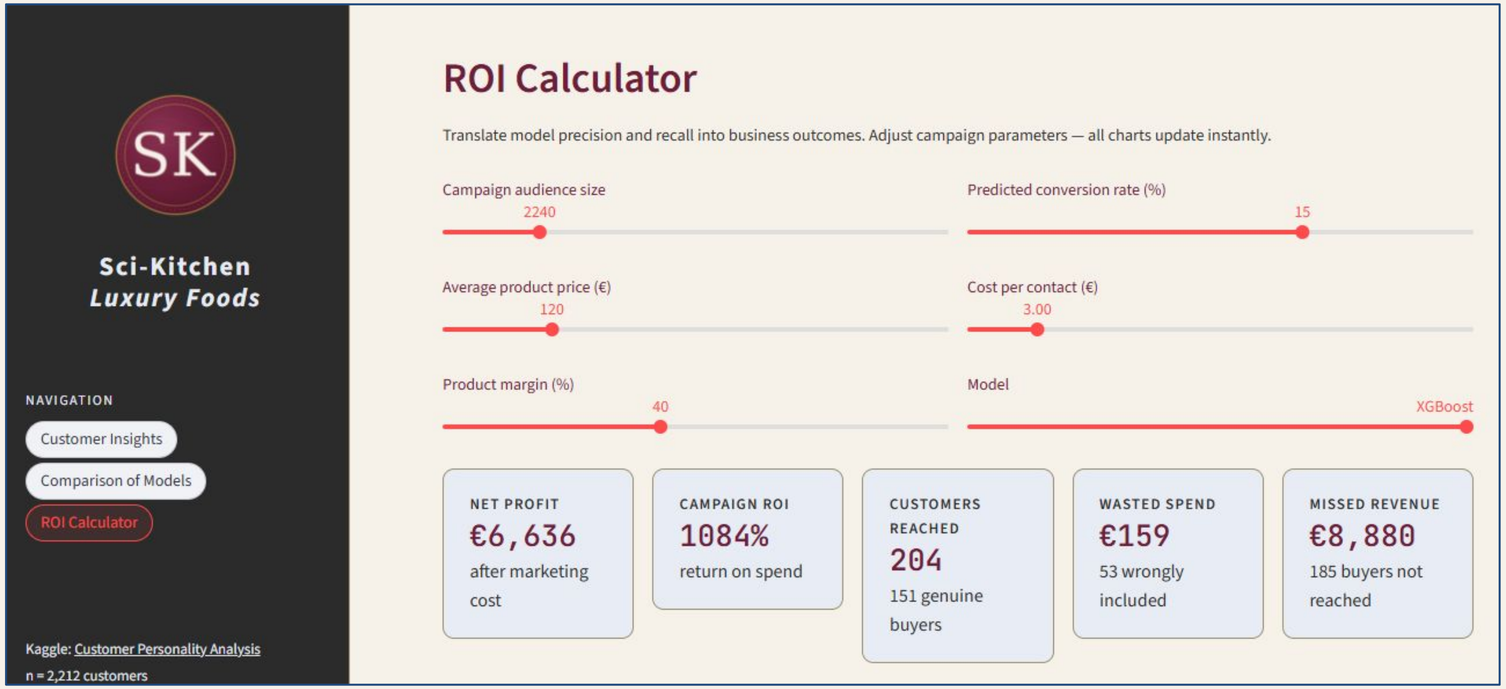


The XGBoost model relied more heavily on segments suggesting a non-linear interaction with campaign behaviour.

Streamlit App to Explore Budget Trade-offs

Deployed a Streamlit app, available here.

- Interactive ROI calculator translating model precision and recall into campaign revenue, wasted spend, and missed profit
- Demographic overview of each customer segment
- Model comparison across AUC, precision and recall



Conclusions & Recommendations

- Digital tracking is the weakest predictor of conversion,.
- CRM data (tenure, recent sales history) combined with segment outperforms demographics and digital behaviour.
    - Prioritise investment in CRM data hygiene and enrichment over digital marketing tools.

- Highest-value segment converts 3x more but doesn't leave a digital footprint.
    - Use demographic features identified in our high-potential high-value segment to build audiences on outbound marketing platforms.
    - Partner with influencers with relevant audiences to bypass poor website engagement.

Key question remains: Where does Miguel discover new promotions?

QUESTIONS?

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Claire Leyden | Ironhack Data Analytics